

Poem as a causal chain

Subject :

No. : 37

Date :

For want of a nail the shoe was lost

" shoe " horse "

" horse " rider "

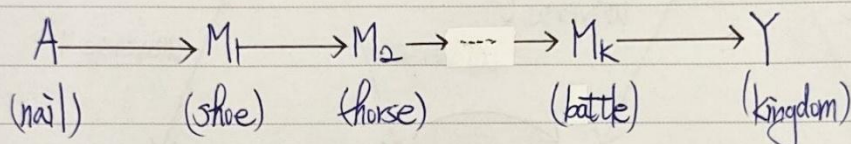
" rider " message "

" message " battle "

" battle " kingdom "

And all for the want of horseshoe nail.

This is exactly the logic of mediation



The influence of A on Y is transmitted through a chain of Mediators.

i) hypothetical intervention on a and define potential mediators and outcomes corresponding to the intervention on a

$$\{M(a), Y(a)\}$$

ii) hypothetical intervention on both a and m and define potential outcomes corresponding to the interventions on a and m

$$\{Y(a, m)\}$$

iii) nested potential outcomes corresponding to intervention on a and $m = M(a^*)$

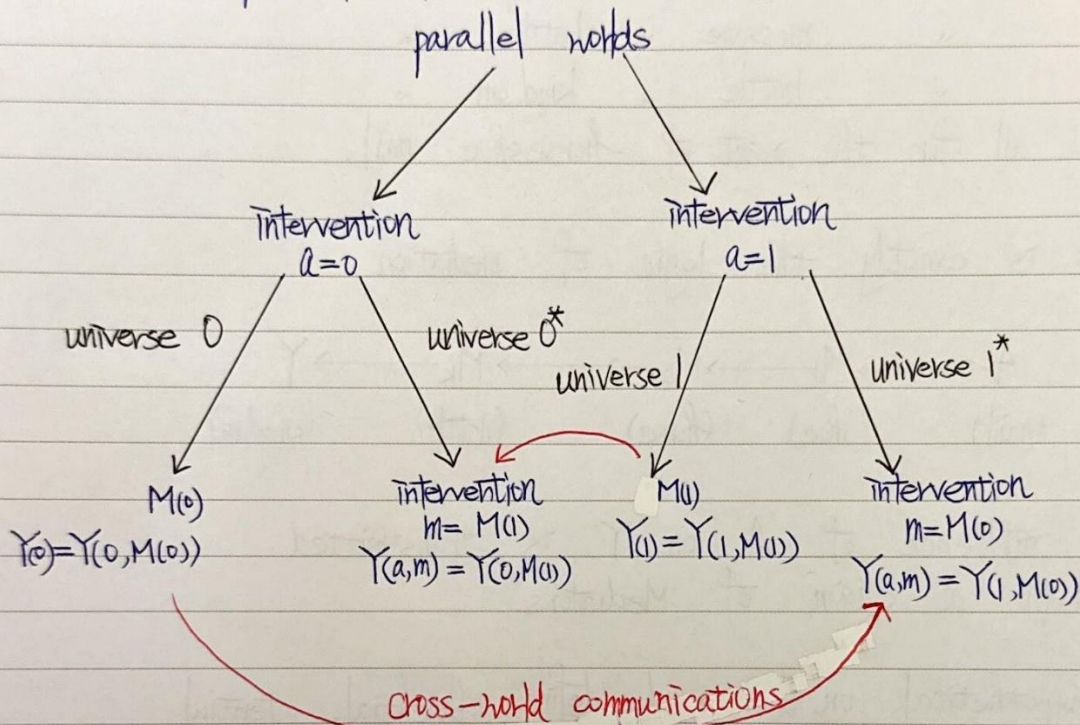
$$\{Y(a, M(a^*))\}$$

$Y(a, M(a^*))$ is the hypothetical outcome if the treatment were set at level a and mediator were set at its

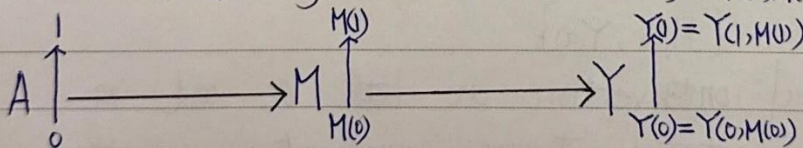
potential level $M(a^*)$ under treatment a^* .

Remark. a and a^* can be different.

★ Cross world potential outcomes



this is the fundamental difference between $\{Y(1), Y(0)\}$ and $\{Y(0, M(1)), Y(1, M(0))\}$



The total effect (TE)

$$\begin{aligned}
 & Y(1) \text{ vs } Y(0) \\
 TE &= E[Y(1) - Y(0)] \quad TE \approx M(0) \rightarrow M(1) \\
 &= E[Y(1, M(1)) - Y(0, M(0))] \\
 & DE \approx 0 \rightarrow 1
 \end{aligned}$$

With counterfactual definitions we can also see why

$$TE = \{E[Y(1, M(1))] - E[Y(1, M(0))]\} + \{E[Y(1, M(0))] - E[Y(0, M(0))]\}$$

$$\equiv NIE + NDE$$

Another decomposition

$$TE = \{E[Y(1, M(1))] - E[Y(0, M(1))]\} + \{E[Y(0, M(1))] - E[Y(0, M(0))]\}$$

$$\equiv TDE + PIE$$

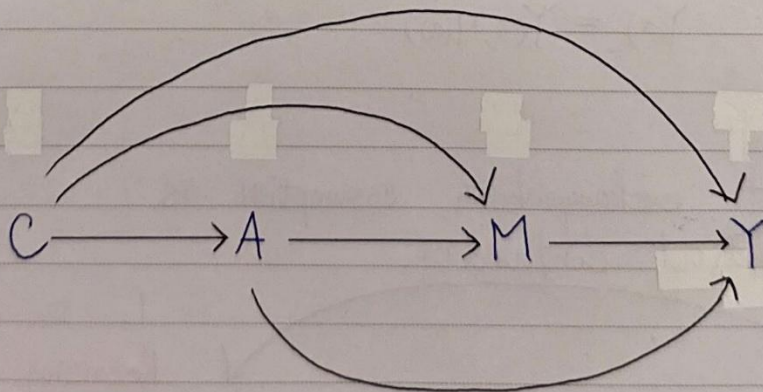
差在 interaction

★ Mediation parameter

$$\phi(a, a^*) = E[Y(a, M(a^*))]$$

then

$$\begin{aligned} TE &= \phi(1, 1) - \phi(0, 0) \\ &= [\phi(1, 1) - \phi(1, 0)] \\ &\quad + [\phi(1, 0) - \phi(0, 0)] \\ &= [\phi(1, 1) - \phi(0, 1)] \\ &\quad + [\phi(0, 1) - \phi(0, 0)] \end{aligned}$$



★ Identification assumption in Mediation

1. Consistency

When $A=a$, the potential outcome $Y(a)$ and $M(a)$ are respectively equal to the observed outcome Y and M

$$\begin{cases} Y(a)=Y \\ M(a)=M \end{cases} \text{ if } A=a$$

When $A=a$ and $M=m$, the potential outcome $Y(a,m)$ is equal to observed outcome Y

$$Y(a,m)=Y \text{ if } \begin{cases} A=a \\ M=m \end{cases}$$

2. Composition

The value of Y that occur if A were set to a

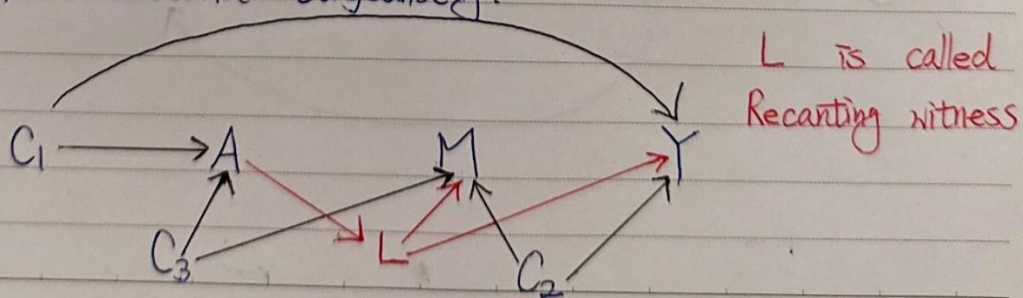
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The value of Y that would occur if A were set to a and M were set to what it would have been if A were set to a

$$Y(a) = Y(a, M(a))$$

3. Exchangeability

A series of exchangeability assumption is crucial to avoid confounding.



Subject :

i) No unmeasured A-Y confounding $\Rightarrow$ Verifiable, RCT
 $Y(a,m) \perp\!\!\!\perp A \mid C$

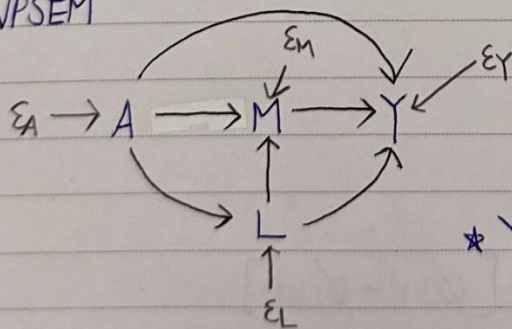
ii) No unmeasured A-M confounding $\Rightarrow$ Verifiable, RCT
 $M(a) \perp\!\!\!\perp A \mid C$

iii) No unmeasured M-Y confounding $\Rightarrow$ It is theoretically verifiable but practically challenging
 $Y(a,m) \perp\!\!\!\perp M \mid A, C$

iv) Crossworld exchangeability $\Rightarrow$ It can't be tested and verified
 $Y(a,m) \perp\!\!\!\perp M(a^*) \mid C$

fail
 when there exist M-Y confounder (L)
 affected by exposure

NPSEM



$$\star Y(a,m) : A=a, M=m$$

$$L(a) = g_L(a, \epsilon_L)$$

$$Y(a,m) = g_Y(a, L(a), m, \epsilon_Y)$$

$$= g_Y(a, g_L(a, \epsilon_L), m, \epsilon_Y)$$

$$= f_Y(\epsilon_L, \epsilon_Y)$$

$$A = g_A(\epsilon_A)$$

$$L = g_L(A, \epsilon_L)$$

$$M = g_M(A, L, \epsilon_M)$$

$$Y = g_Y(A, L, M, \epsilon_Y)$$

$$\star M(a^*) : A=a^*, L(a^*) = g_L(a^*, \epsilon_L)$$

$$M(a^*) = g_M(a, L(a^*), \epsilon_M) = g_M(a, g_L(a^*, \epsilon_L), \epsilon_M)$$

$$= f_M(\epsilon_L, \epsilon_M)$$

Double A

★ Mediation formula

$$E[Y(a, M(a^*))] = \int_C E[Y(a, M(a^*)) | C=c] P(C=c)$$

$$= \int_C \left\{ \int_m E[Y(a, M(a^*)) | M(a^*)=m, C=c] P(M(a^*)=m | C=c) \right\} P(C=c)$$

$$(\text{consistency}) = \int_C \int_m E[Y(a, m) | M(a^*)=m, C=c] P(M(a^*)=m | C=c) P(C=c)$$

$$(Y(a, m) \perp\!\!\!\perp M(a^*) | C) = \int_C \int_m E[Y(a, m) | C=c] P(M(a^*)=m | C=c) P(C=c)$$

$$(Y(a, m) \perp\!\!\!\perp A | C) = \int_C \int_m E[Y(a, m) | A=a, C=c] P(M(a^*)=m | C=c) P(C=c)$$

$$(Y(a, m) \perp\!\!\!\perp M | A, C) = \int_C \int_m E[Y(a, m) | M=m, A=a, C=c] P(M(a^*)=m | C=c) P(C=c)$$

$$(\text{consistency}) = \int_C \int_m E[Y | M=m, A=a, C=c] P(M(a^*)=m | C=c) P(C=c)$$

$$(M(a^*) \perp\!\!\!\perp A | C) = \int_C \int_m E[Y | M=m, A=a, C=c] P(M(a^*)=m | A=a^*, C=c) P(C=c)$$

$$(\text{consistency}) = \int_C \int_m E[Y | M=m, A=a, C=c] P(M=m | A=a^*, C=c) P(C=c)$$

$$= Q(a, a^*)$$

$$\phi(a, a^*) \stackrel{\Delta}{=} Q(a, a^*)$$

identify

$$TE = NDE + NIE$$

$$\hat{\phi}(1,1) - \hat{\phi}(0,0) = [\hat{\phi}(1,0) - \hat{\phi}(0,0)] + [\hat{\phi}(1,1) - \hat{\phi}(1,0)]$$

✍ Estimation

Two traditional approaches to mediation analysis

i) The Difference Method

ii) The Product Method

- ★ The difference method has been employed more frequently in epidemiology and biomedical sciences. It consists of two regression models.

$$Y = \beta_0 + \beta_1 A + \varepsilon_1$$

$$Y = \theta_0 + \theta_1 A + \theta_2 M + \varepsilon_2$$

The exposure coef θ_1 , in the model that includes the mediator is then generally take as a measure of DE

$$\begin{aligned} DE &= \theta_1 \\ TE &= \beta_1 \end{aligned} \Rightarrow IE = \beta_1 - \theta_1$$

- ★ A slightly different method, sometimes called the product-of-coefficients method, is used w/ more frequency in the social sciences.

Two regressions are employed.

$$\begin{aligned} Y &= \theta_0 + \theta_1 A + \theta_2 M + \varepsilon_2 \\ M &= \alpha_0 + \alpha_1 A + \varepsilon_3 \end{aligned}$$

the product $\alpha_1 \theta_2$, taken as a measure of indirect effect

$$IE = \alpha_1 \theta_2$$

- ★ Fortunately, for $\begin{cases} Y: \text{conti} \\ M: \text{conti} \end{cases}$ w/ linear regression fit by OLS

the two approaches are coincide.

$$\beta_1 - \theta_1 = \alpha_1 \theta_2$$

Recall the identification of $\phi(a, a^*) = E[Y(a, M(a^*))]$

$$\int_c \int_m E(Y | M=m, A=a, C=c) P(M=m | A=a^*, C=c) P(C=c)$$

$$\text{or } \int_c \int_m E(Y | M=m, A=a, C=c) f_{M|A,C}(m | a^*, c) f_C(c) dm dc$$

(i) Model Y:

$$E(Y|A, M, C) = \theta_0 + \theta_1 A + \theta_2 M + \theta_c C$$

(ii) Model M:

$$E(M|A, C) = \alpha_0 + \alpha_1 A + \alpha_c C$$

$$\Rightarrow Q(a, \bar{a}) = \int_c \int_m (\theta_0 + \theta_1 a + \theta_2 m + \theta_c C) f_{M|A, C}(m|a^*, c) f_c(c) dm dc$$

$$= \theta_0 + \theta_1 a + \theta_2 \int_c \int_m m f_{M|A, C}(m|a^*, c) dm f_c(c) dc + \theta_c E(C)$$

$$B = \theta_2 \int_c E(M|A=a^*, C=c) f_c(c) dc$$

$$= \theta_2 \int_c (\alpha_0 + \alpha_1 a^* + \alpha_c C) f_c(c) dc$$

$$= \theta_2 (\alpha_0 + \alpha_1 a^* + \alpha_c E(C))$$

then

$$Q(a, \bar{a}) = \theta_0 + \theta_1 a + \theta_2 (\alpha_0 + \alpha_1 a^* + \alpha_c E(C)) + \theta_c E(C)$$

$$\theta_1 a + \theta_2 \alpha_1 a^* + \text{Const}$$

$$NDE = \phi(1, 0) - \phi(0, 0) = Q(1, 0) - Q(0, 0) = \theta_1 \rightarrow \text{DE in difference method}$$

$$NIE = \phi(1, 1) - \phi(1, 0) = Q(1, 1) - Q(1, 0) = \theta_2 \alpha_1$$

$\rightarrow$ IE in product method

$$(iii) E(Y|A, C) = \theta_0 + \theta_1 A + \theta_c C$$

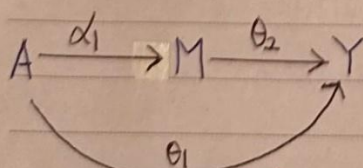
$$\Downarrow$$

$$E_M(E(Y|M, A, C)) = E_M(\theta_0 + \theta_1 A + \theta_2 M + \theta_c C)$$

$$= \theta_0 + \theta_1 A + \theta_2 E_M(M|A, C) + \theta_c C$$

$$= \theta_0 + \theta_1 A + \theta_2 (\alpha_0 + \alpha_1 A + \alpha_c C) + \theta_c C$$

$$= (\theta_0 + \theta_2 \alpha_0) + (\theta_1 + \theta_2 \alpha_1) A + (\theta_c + \theta_2 \alpha_c) C$$



Difference Method

$$\theta_1 = \theta_1 + \theta_2 \alpha_1$$

Product Method

★ Exposure - Mediator Interaction

$$\text{Model } Y : E(Y|M, A, C) = \theta_0 + \theta_1 A + \theta_2 M + \theta_3 \underbrace{AM}_{\substack{\text{A-M} \\ \text{interaction}}} + \theta_4 C$$

Recall

$$Q(a, a^*) = \int_C \int_m E(Y|M=m, A=a, C=c) f_{M|A,C}(m|a^*, c) f_C(c) dm dc$$

$$= \theta_0 + \theta_1 a + \underbrace{\int_C \int_m (\theta_2 + \theta_3 a) m f_{M|A,C}(m|a^*, c) f_C(c) dm dc}_{\substack{\parallel \\ (\Delta)}} + \theta_4 E(C)$$

$$(\Delta) = (\theta_2 + \theta_3 a) \int_C E(M|A=a^*, C=c) f_C(c) dc$$

$$= (\theta_2 + \theta_3 a) E(\alpha_0 + \alpha_1 a^* + \alpha_2 C)$$

$$= (\theta_2 + \theta_3 a) [\alpha_0 + \alpha_1 a^* + \alpha_2 E(C)]$$

then

$$Q(a, a^*) = (\theta_0 + \theta_1 a) + (\theta_2 + \theta_3 a) [\alpha_0 + \alpha_1 a^* + \alpha_2 E(C)] + \theta_4 E(C)$$

$$= \theta_0 + \theta_2 \alpha_0 + \theta_2 \alpha_2 E(C) + \theta_4 E(C)$$

$$+ (\theta_1 + \alpha_0 \theta_3 + \alpha_2 \theta_3 E(C)) a$$

$$+ \theta_2 \alpha_1 a^*$$

$$+ \theta_3 \alpha_1 a a^*$$

$$\triangleq \text{Const} + Q_1 a + Q_2 a^* + Q_3 a a^*$$

$$NIE = Q(1, 1) - Q(1, 0) = (Q_1 + Q_2 + Q_3) - Q_1 = \alpha_1 (\theta_2 + \theta_3)$$

$$NDE = Q(1, 0) - Q(0, 0) = Q_1 = \theta_1 + \theta_3 (\alpha_0 + \alpha_2 E(C))$$

★ MC Estimation

$$Q(a, a^*) = \int_C \int_m E(Y|M=m, A=a, C=c) P(M=m|A=a^*, C=c) P(C)$$

$$= \frac{1}{n} \sum_{i=1}^n \hat{Y}_i$$

$$E(Y|M=m, A=a, C=c) = \theta_0 + \theta_1 a + \theta_2 m + \theta_4 C = g_Y(a, m, C; \theta)$$

(+ $\theta_3 am + \theta_4 C + \theta_5 mc$ if consider interaction)

$$\hat{Y}_i = g_Y(A_i, \hat{M}_i, \hat{C}_i; \hat{\theta})$$

$$\hat{C}_i \sim P(C=c)$$

$$\hat{M}_i \sim \hat{P}(M=m|A=a^*, \hat{C}_i)$$

Id C_i A_i M_i Y_i 1. Build model for Y, M

$$E(Y|M, A, C) = g_Y(A, M, C, \theta)$$

$$E(M|A, C) = g_M(A, C, \alpha)$$

2. $\hat{\alpha}, \hat{\Delta M}, \hat{\theta}$

by REML

(D1) id C A M $Y \leftarrow Y(1) = Y(1, M(1))$

1
2
...
n

$$C_i \sim \hat{P}(C=c), A=1$$

$$M \sim g_M(A=1, C, \hat{\alpha}, \hat{\Delta M})$$

$$Y \sim g_Y(A=1, M, C, \hat{\theta})$$

$$\Rightarrow \frac{\sum Y_i}{n} = \hat{E}(Y(1)) = \hat{\phi}(1, 1)$$

(D2) id C A M $Y \leftarrow Y(0) = Y(0, M(0))$

1
2
...
n

$$C_i \sim \hat{P}(C=c), A=0$$

$$M \sim g_M(A=0, C, \hat{\alpha}, \hat{\Delta M})$$

$$Y \sim g_Y(A=0, M, C, \hat{\theta})$$

$$\Rightarrow \frac{\sum Y_i}{n} = \hat{E}(Y(0)) = \hat{\phi}(0, 0)$$

(D3) id C A M $Y \leftarrow Y(1, M(0))$

1
2
...
n

$$C_i \sim P(C=c), A=1$$

$$Y \sim g_Y(A=1, M, C, \hat{\theta})$$

$$\Rightarrow \frac{\sum Y_i}{n} = \hat{E}(Y(1, M(0))) = \hat{\phi}(1, 0)$$

統計

流行病學

JASA, JRRSB, Biometrika, Annals $\Leftrightarrow$ Epidemiology, AJE, IJE, EJE